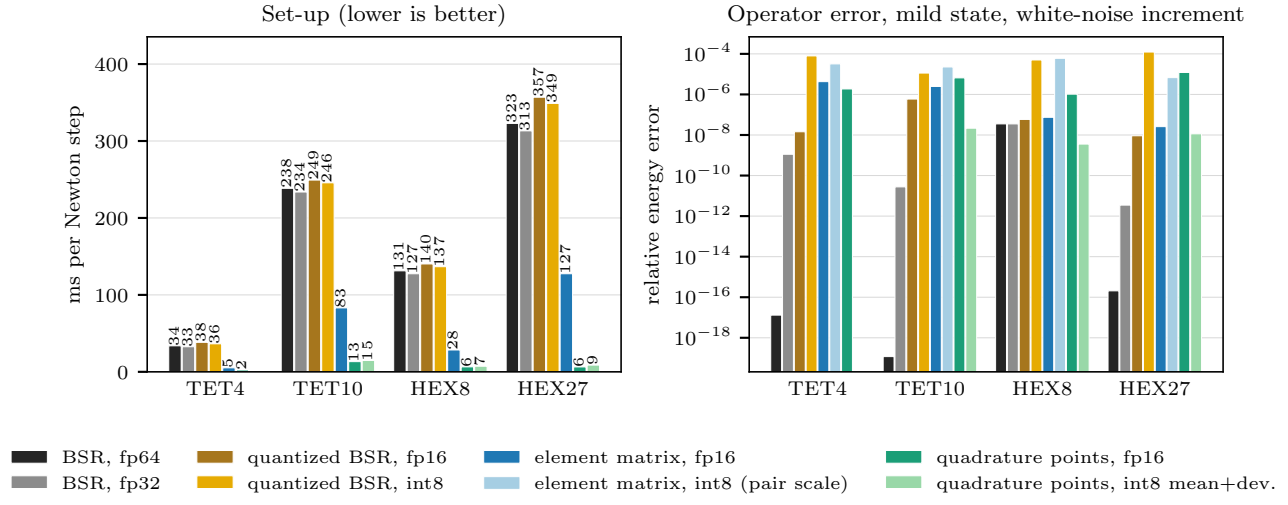

Set-up (lower is better)
ms per Newton step
0
100
200
300
400
34
33
38
36
5
2
238
234
249
246
83
13
15
131
127
140
137
28
6
7
323
313
357
349
127
6
9
TET4
TET10
HEX8
HEX27
Operator error, mild state, white-noise increment
relative energy error
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{-12}$
$10^{-14}$
$10^{-16}$
$10^{-18}$
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.